KOTARO AMAYA

Ph.D. Researcher | Humanoid Learning, Human Motion, Embodied AI

Email | Website | Google Scholar | GitHub | LinkedIn

RESEARCH FOCUS

I build systems that learn interaction structure from human motion and transfer it across embodiments into physically executable humanoid behaviors, spanning generative modeling, reinforcement learning, simulation, and real-robot deployment.

RESEARCH EXPERIENCE

Carnegie Mellon University, Pittsburgh, USA | Visiting Scholar, Machine Learning Department — Oct 2025 – Oct 2026

- Humanoid learning for human–robot physical interaction on Unitree G1: contact-aware human-to-robot retargeting, reinforcement-learning policy training, depth-conditioned visual policies, and real-robot deployment.
- Learning task-specific pose, contact, and temporal constraints from a few human motion clips as priors for guided motion generation and humanoid skill learning without per-skill reward engineering.
- Built robotics research infrastructure across Isaac Lab, MuJoCo/MJWarp, Blender, ONNX, and SLURM-based GPU clusters.

Keio University, Yokohama, Japan | Ph.D. Research, Isogawa Lab — Apr 2025 – Present

- Long-horizon, text-driven human motion generation with temporal consistency across successive instructions, in collaboration with Sony AI.

Keio University, Yokohama, Japan | M.E. Research, Isogawa Lab — Apr 2023 – Mar 2025

- Federated learning for privacy-preserving mmWave-based 3D human mesh estimation under heterogeneous client distributions.

Keio University, Yokohama, Japan | B.E. Research, Isogawa Lab — Apr 2022 – Mar 2023

- mmWave-based 3D human pose and mesh estimation robust to diverse, non-upright poses; built the lab's motion-capture and mmWave sensing setup.

EDUCATION

Keio University | Ph.D. in Computer Science — Expected Mar 2028

Keio University | M.E. in Computer Science — Mar 2025

Keio University | B.E. in Computer Science — Mar 2023

PUBLICATIONS

First-authored — International

[1] Y. Shibata*, K. Amaya*, K. Yamazaki, L. Jayanti, Y. Aoki, M. Isogawa, and K. Fragkiadaki. "VLM-Guided Contact Retargeting and Generative Partner Modeling for Humanoid–Human Interaction." Conference on Robot Learning (CoRL), 2026. Acceptance rate 33%; *equal contribution.

[2] K. Amaya and M. Isogawa. "Adaptive and Robust mmWave-Based 3D Human Mesh Estimation for Diverse Poses." IEEE ISMAR Adjunct, IW-P3HSE Workshop, 2024. Oral presentation.

[3] K. Amaya and M. Isogawa. "Adaptive and Robust mmWave-Based 3D Human Mesh Estimation for Diverse Poses." IEEE ICIP, 2023. Oral presentation (top 14.5%).

First-authored — Domestic (Japan)

[4] Y. Shibata*, K. Amaya*, K. Yamazaki, L. Jayanti, Y. Aoki, M. Isogawa, and K. Fragkiadaki. "VLM-Guided Contact Retargeting and Generative Partner Modeling for Deployable Humanoid–Human Interaction." MIRU (Meeting on Image Recognition and Understanding), 2026; in Japanese; *equal contribution.

[5] K. Amaya*, S. Kato*, T. Shibuya, J. Tanke, K. Fukuda, Y. Mitsufuji, and M. Isogawa. "Text Guided Long-Term Human Motion Generation with Temporal Consistency." MIRU, 2025. Peer-reviewed; in Japanese; *equal contribution.

[6] K. Amaya and M. Isogawa. "Improving Noise Robustness of mmWave-Based Human Pose Estimation by Extracting Human Regions from Point Clouds." MIRU, 2023. Peer-reviewed short oral (top 46%); in Japanese.

Co-authored — International

[7] S. Kato, D. Yang, K. Amaya, Y. Wada, K. Fukuda, T. Shibuya, Y. Mitsufuji, and M. Isogawa. "ELMA: Benchmarking Anaphoric Compositions for Long-Term Text-to-Motion Generation." NeurIPS Evaluations & Datasets Track, 2026.

[8] H. Deguchi, R. Hori, K. Amaya, T. Maruyama, M. Tada, and H. Saito. "Map-Mono-Ego: Map-Grounded Global Human Pose Estimation from Monocular Egocentric Video." IEEE ICIP, 2026; also ECCV Workshop on Human-Motion World Models (HuMoWM), 2026.

Co-authored — Domestic (Japan)

[9] D. Yang, S. Kato, K. Amaya, Y. Wada, K. Fukuda, T. Shibuya, Y. Mitsufuji, and M. Isogawa. "ACE: Anaphoric Composition Evaluation for Long-Term Text-to-Motion Generation." MIRU, 2026.

[10] S. Kato, D. Yang, K. Amaya, Y. Wada, K. Fukuda, T. Shibuya, Y. Mitsufuji, and M. Isogawa. "Benchmarking Cross Dependencies for Long-Term Text-to-Motion Generation." MIRU, 2026.

INDUSTRY EXPERIENCE

Sony — R&D Intern | Explored sensing approaches for cognitive-load estimation. — Aug–Sep 2023

CyberAgent — ML Engineer Intern | Worked on applied machine-learning engineering. — Jul–Aug 2023

OMRON SINIC X — Research Intern | Studied representation learning for materials informatics. — Mar 2023

TECHNICAL SKILLS & FUNDING

Programming: Python | **ML/Robotics:** PyTorch, Isaac Lab, MuJoCo/MJWarp, Unitree G1, ONNX | **Tools:** Blender, Git, Linux, SLURM

Funding: JST BOOST Fellow (2025–2027)